

IFS-SED: Incremental Few-Shot Sound Event Detection Using Explicit Learning and Calibration

Ming Feng
Tongji University
Shanghai, China
andrew_tal@163.com

Kele Xu*
National University of Defense
Technology
Changsha, China
xukelele@163.com

Hengxing Cai
4Paradigm
Beijing, China
1220176667@qq.com

ABSTRACT

Sound event detection (SED) refers to recognizing the sound events in a continuous audio signal, which has drawn increasing interest during recent decades. The applications of SED seem to be evident in many fields, ranging from surveillance to monitoring applications. Despite the sustainable efforts that have been made, most of the previous attempts are performed on the closed-set, as only fixed and known sound event classes can be employed during the training. In this paper, we present our incremental few-shot SED framework under the open-set settings, as a practical machine listening system should be able to address unknown sound events. Specifically, an explicit learning and calibration-based multi-stage learning framework is utilized to address the challenges of catastrophic forgetting, and aim to achieve a better trade-off between stability and plasticity. To compress the model efficiently, the model prune and self-distillation paradigm are combined used for the model compression, thus our system can be deployed for the resource-limited devices. Our framework can also provide an uncertain estimation for the inference. Lastly, an interactive interface is presented to demonstrate the functions of our system.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

sound event detection, incremental learning

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1 INTRODUCTION

Sound signals can be found in our daily life, and the interest in automated detecting the sound events from the signals has surged in recent years [11, 12, 18]. The applications of SED seem to be

evident for a variety of domains, such as multimedia search and retrieval [2], sound monitoring and audio caption [7]. Sustainable efforts have been made for the sound event detection (SED) task, leveraging the shallow-architecture classifier, the deep neural networks [14] and Transformer architectures [4, 5]. The detection task can be very challenging due to the vastness of the sounds signals, or the contamination of noise. Moreover, sound events can be overlapped with multiple other events, which dramatically increased the difficulties of this task. Some attempts also been made to address the classes imbalance problem and the variations of data quality (such as the noisy label [19]), for the detection task.

While successful, two main challenges are confronted by the practical SED systems: their closed-set nature and limited labeled data (especially for the rare sound events). Firstly, the detection imposes a fixed class vocabulary and can only be performed for the known class that was included in the training data. However, a practical SED system is an open-set problem in nature and the system should also be able to deal with unknown sound events as well. It is also required the model to remember the previously learned sound events after adopting the new events, and this kind of learning ability can be referred to as incremental learning / continual learning (or lifelong learning).

Moreover, most previous supervised learning-based approaches require large-number labeled datasets, which is of great difficulty to collect and annotate the exhaustive training samples with different sound events. Real-world applications have very few labeled sound events for incremental learning. Thus, it is important for the SED model to be capable of incrementally learning sound events from very few training examples [9, 13]. Compared with the incremental learning with a large number of labeled datasets, the goal of our SED system is more challenging, since the number of new training samples for the sound events is very limited [16].

In this demo, we explore to build an incremental learning framework for SED tasks on resources-limited devices, such as mobile phones. Specifically, several key modules have been provided to detect the sound event for the audio signals, including incremental learning under the few-shot settings, uncertainty estimation, model prune, and self-distillation.

2 INCREMENTAL FEW-SHOT SED SYSTEM

The standard SED system aims to predict the labels and the corresponding onset/offset time. Here, unlike standard SED, incremental few-shot SED system aims to detect the novel sound events categories from only a few labeled data points, while not forgetting the base categories on which it was initially trained, which is a core step towards achieving general machine listening [15].

*Corresponding author

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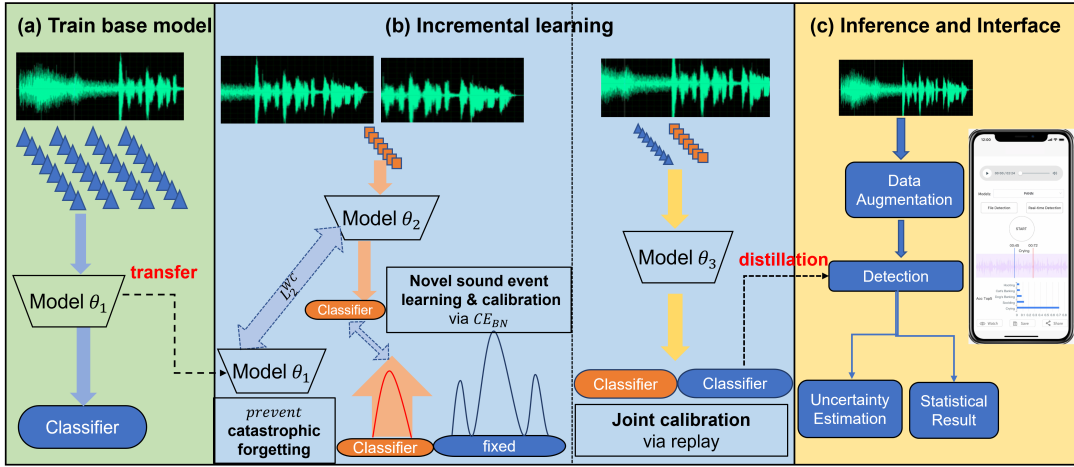


Figure 1: Framework of proposed approach. (a) presents the offline training stage for the base model, while the few-shot incremental learning part is given in (b) for the sound event detection task. (c) provides the inference part and snapshot for the interface deployed on the resources-limited devices.

2.1 Overall design

Figure 1 illustrates an overview framework of our work, which can be divided into three main modules. In the first part, offline training for the detection task can be conducted. While, in the second part, incremental learning can be performed, leveraging a multi-stage-based framework with few-shot scenarios. In the final step, the inference can be conducted using the compressed model and uncertainty estimation. The technical details can be found in the following sections. The snapshot of the interactive interface is also given in the figure.

2.2 Incremental Few-shot SED

In our practical implementation, we aim to address three main challenges for the Incremental Few-shot SED task, including the incremental learning new classes by using only a few samples per sound event, preventing catastrophic forgetting of the base sound event, and the calibration across old and new sound event [1, 10]. Specifically, we endeavor to achieve the best performance on both the base and novel sound event detection simultaneously.

To achieve balanced performance, we address different phases of our multi-stage framework. In the first section, we pre-train the detection model on base sound events with abundant data. In the second phase we use L_2 weight constraints to preserve knowledge, thus constraining the change of the new model. This phase is the most important part to obtain strong performance for both novel and base sound events. In the final stage, we aim to calibrate the incremental SED performance, leveraging the balanced replay of novel and base sound events. The balanced replay is a common strategy in incremental learning.

2.3 Uncertainty Estimation

Deep neural networks are prone to provide overconfident and miscalibrated predictions [6], which can be harmful or offensive in practical settings, especially for rare sound events. Proper uncertainty quantification is crucial for audio signal analysis[17], which

is however under-explored in previous studies. For most of the previous attempts for SED tasks leveraging deep learning, the deep model lacks a representation of uncertainty. In this demo, except for the ability for incremental learning, we also equip our model with the uncertainty estimation ability, using *Monte Carlo Dropout mechanism* and *Deep Ensemble*.

2.4 Model Compression and Serving

Generally, a SED model can consist of tens of millions of learned parameters, which cannot be employable for resource-limited devices [3] directly, such as mobile phones. Thus, the model compression step is one necessary step toward the SED systems. In our implementation, both the model prune and self-Knowledge distillation [8] strategy are combined used to compress the model size efficiently. After the multi-stage-based training, both the large (teacher) model and small (student) model are uploaded to a cloud server automatically, which is equipped with the two NVIDIA GTX 3090Ti GPUs, thus the inference can be conducted in almost real-time. Using the presented interface, the user can upload the audio recordings for the evaluation. The detection can be conducted using the compact model stored in the mobile devices, or the large model which is stored in the cloud server. We also provide an interactive interface, to facilitate the user to label the novel sound events, which can further be unitized for the learning phase.

3 CONCLUSION

In this study, we propose a few-shot incremental learning framework for SED tasks, which is a long-standing, important function for practical machine listening. Here, we aim to demonstrate a multi-stage SED framework that address the challenges of catastrophic forgetting and aims to achieve a better trade-off between stability and plasticity. Moreover, an interface is provided to visualize the detection task, equipped with the uncertainty estimation ability. We hope our tool can be helpful for a practical machine listening system.

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